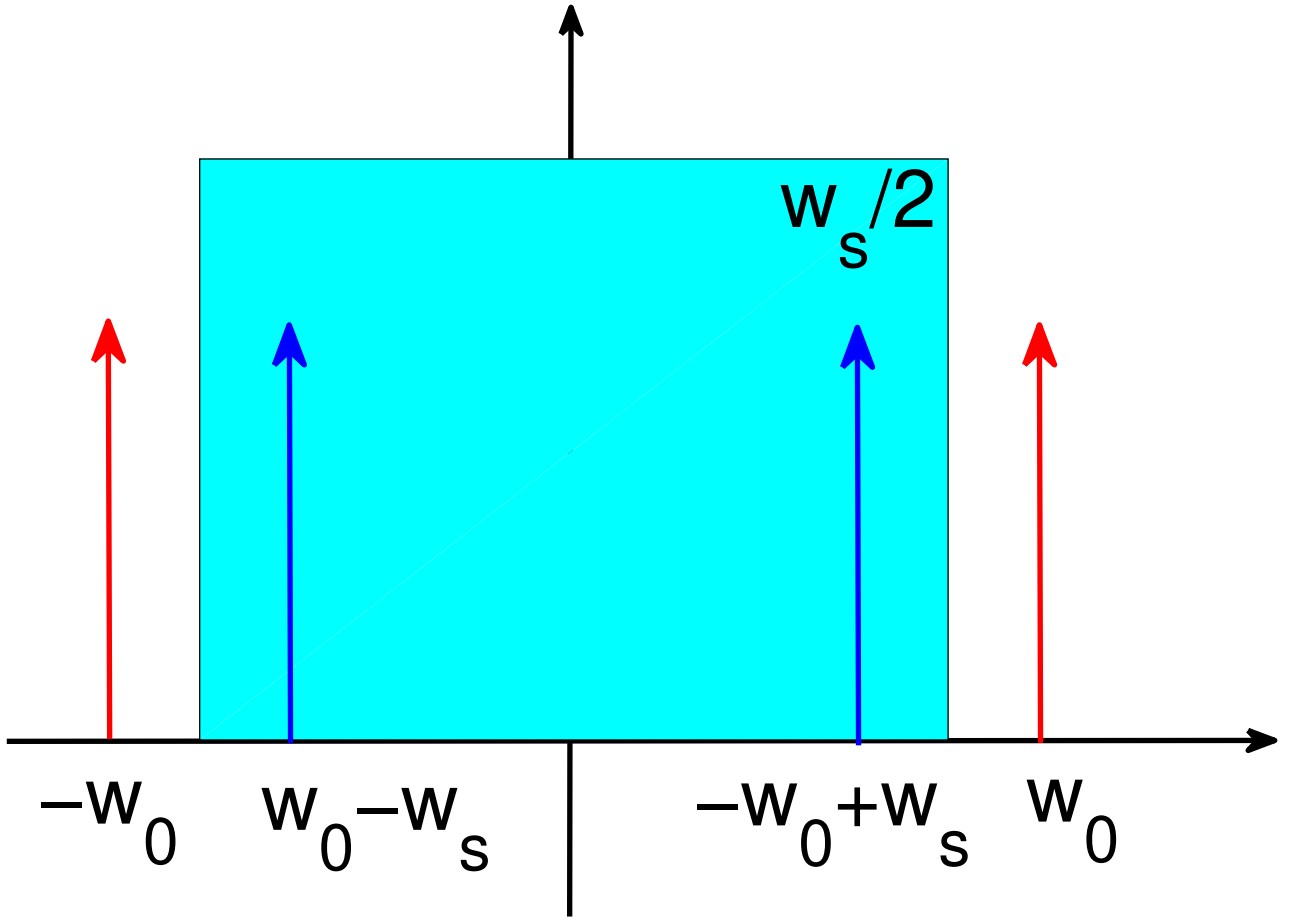
$w_s/2$
$-w_0$
w_0-w_s
$-w_0+w_s$
w_0